# Unit 1.9 Matrices

## Paper 1: Intermediate

**1.** Cheng and Xin are each tiling a bathroom.

Cheng needs 5 packs of floor tiles, 10 packs of wall tiles, 4 bags of adhesive and 2 bags of grout.

Xin needs $x$ packs of floor tiles, $(x + 2)$ packs of wall tiles, 2 bags of adhesive and 1 bag of grout.

This information can be represented by the matrix $\mathbf{P} = \begin{pmatrix} 5 & 10 & 4 & 2 \\ x & x+2 & 2 & 1 \end{pmatrix}$.

In a store, a pack of floor tiles costs \$140, a pack of wall tiles \$105, a bag of adhesive \$13 and a bag of grout \$9.

From an online supplier, a pack of floor tiles costs \$150, a pack of wall tiles \$100, a bag of adhesive \$12 and a bag of grout \$8.

This information can be represented by the matrix $\mathbf{Q} = \begin{pmatrix} 140 & 150 \\ 105 & 100 \\ 13 & 12 \\ 9 & 8 \end{pmatrix}$.

**(a)** Find, in terms of $x$, the matrix $\mathbf{T} = \mathbf{PQ}$. [2]

**(b)** Explain what the elements of the first row of matrix $\mathbf{T}$ represent. [1]

**(c)** The online price for Xin's order is \$2 more than the price in the store.
Find $x$. [2]

**(d)** Xin orders all her items from the store or she orders all her items from the online supplier.
Find the lowest price she could pay for all her items. [1]

[N22/I/18]

**2.** In garden centre A, a bag of soil costs \$36, a trough costs \$40 and a pot costs \$68.

The same items can be bought in garden centre B.

In garden centre B the soil costs \$$x$, the trough \$39 and the pot \$65.

This information can be represented by the matrix $\mathbf{N} = \begin{array}{c} \begin{array}{cc} \text{A} & \text{B} \end{array} \\ \begin{pmatrix} 36 & x \\ 40 & 39 \\ 68 & 65 \end{pmatrix} \end{array} \begin{array}{c} \\ \text{S} \\ \text{T} \\ \text{P} \end{array}$

**(a)** Tom buys 5 bags of soil, 4 troughs and 2 pots.
Lim buys 6 bags of soil, 3 troughs and $y$ pots.
Represent this information in a $2 \times 3$ matrix $\mathbf{M}$. [1]

**(b)** Find, in terms of $x$ and $y$, the matrix $\mathbf{P} = \mathbf{MN}$. [2]

**(c)** State what the first element of matrix $\mathbf{P}$ represents. [1]

**(d)** The total cost of Tom's purchases would be the same in both garden centres.
Find the value of $x$. [1]

**(e)** Lim would save \$3 by buying her purchases in garden centre B.
Find the value of $y$. [1]

[N21/I/22]

**3.** The cost, in cents, of printing black and white copies and colour copies on two printers, A and B, is represented by the matrix **P**.

$$\mathbf{P} = \begin{array}{c} \\ \\ \end{array}\begin{array}{cc} \text{A} & \text{B} \end{array}$$
$$\mathbf{P} = \begin{pmatrix} 10 & 8 \\ 40 & 45 \end{pmatrix} \begin{array}{l} \text{B/W} \\ \text{C} \end{array}$$

**(a)** Koh, Min and Suman each have access to the printers A and B.

Koh needs to print 24 black and white copies and 14 colour copies.

Min needs to print 29 black and white copies and $x$ colour copies.

Suman needs to print 20 black and white copies and 15 colour copies.

Represent this information in a $3 \times 2$ matrix **Q**. [1]

**(b)** Find, in terms of $x$, the matrix $\mathbf{T} = \mathbf{QP}$. [2]

**(c)** If Koh and Min both use printer A, Min would pay 70 cents less than Koh.

Find the value of $x$. [1]

[N19/I/21]

**4.** A tennis club charges by the hour for the hire of its courts.

In one week, courts are hired for 28 hours by children (C), for 170 hours by adults (A) and for 95 hours by senior citizens (S).

The next week the courts are hired for $x$ hours by children, for 176 hours by adults and for 89 hours by senior citizens.

**(a)** Represent this information in a $2 \times 3$ matrix, **P**. [1]

**(b)** The charge for one hour's hire is \$3 for children, \$5 for adults and \$4 for senior citizens.

Find, in terms of $x$, the matrix $\mathbf{R} = \mathbf{P}\begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix}$. [2]

**(c)** Explain what each element in matrix **R** represents. [1]

**(d)** The total amount of money collected for the hire of the courts was the same for both weeks.

Calculate $x$. [1]

[N17/I/19]

**5.** In supermarket A, water costs \$1.50 per litre, milk costs \$2.40 per litre and cola costs \$1.40 per litre.

In supermarket B, water costs \$0.20 more per litre, milk costs \$0.40 less per litre and cola costs \$0.10 less per litre.

This information can be represented by the matrix $\mathbf{Q} = \begin{array}{c} \begin{array}{cc} \text{A} & \text{B} \end{array} \\ \begin{pmatrix} 1.5 & 0.2 \\ 2.4 & -0.4 \\ 1.4 & -0.1 \end{pmatrix} \end{array} \begin{array}{l} \\ \text{W} \\ \text{M} \\ \text{C} \end{array}$.

**(a)** Natalie and Hadi go shopping.

Natalie buys 4 litres of water, 2 litres of milk and 3 litres of cola.

Hadi buys 3 litres of water and 4 litres of cola.

Represent their purchases in a $2 \times 3$ matrix **P**. [1]

**(b)** Evaluate the matrix $\mathbf{R} = \mathbf{PQ}$. [2]

**(c)** How much money would Hadi save by shopping in supermarket A? [1]

**(d)** Natalie shops in supermarket B.

She has a shopping voucher giving a discount of 10%.

How much does she pay altogether for her items? [2]

[N15/I/22]

1. A holiday club is open for a morning session and an afternoon session for 5 days each week.
Each child attends every day for either the morning session or the afternoon session.
The children attending are split into three groups, P, Q and R.
The matrix **H** shows the number of children in each group in week 1.

$$\begin{array}{cc} & \begin{array}{ccc} P & Q & R \end{array} \\ \mathbf{H} = & \begin{pmatrix} 10 & 12 & 16 \\ 14 & 16 & 20 \end{pmatrix} \begin{array}{l} \text{morning} \\ \text{afternoon} \end{array} \end{array}$$

**(a)** The matrix **B** shows the number of boys in each group.

$$\begin{array}{cc} & \begin{array}{ccc} P & Q & R \end{array} \\ \mathbf{B} = & \begin{pmatrix} 7 & 6 & 8 \\ 6 & 9 & 7 \end{pmatrix} \begin{array}{l} \text{morning} \\ \text{afternoon} \end{array} \end{array}$$

Find the number of girls attending the afternoon session of group Q. [1]

**(b)** Each child is charged a fee for each session.
The session fee is \$30 for group P, \$26 for group Q and \$24 for group R.
Represent the fees in a $3 \times 1$ column matrix **F**. [1]

**(c)** Evaluate the matrix $\mathbf{M} = \mathbf{HF}$. [2]

**(d)** State what each element of matrix **M** represents. [1]

**(e)** Calculate the total amount taken in fees by the club in one week. [1]

**(f)** In week 2, the number of children attending the club changes.
The number of children attending group P increases by 50%.
The number of children attending group Q increases by 50%.
The number of children attending group R decreases by 25%.
Calculate the percentage change in the amount taken in fees from week 1 to week 2. [3]

[N20/II/2]

2. A bakery makes fruit pies.
The bakery operates for 7 days a week.
The matrix, **M**, shows the number of pies of different types that are made each day.

$$\begin{array}{cc} & \begin{array}{ccc} \text{Small} & \text{Medium} & \text{Large} \end{array} \\ \mathbf{M} = & \begin{pmatrix} 70 & 40 & 30 \\ 50 & 30 & 30 \end{pmatrix} \begin{array}{l} \text{Apple} \\ \text{Cherry} \end{array} \end{array}$$

**(a)** Evaluate the matrix $\mathbf{P} = 7\mathbf{M}$. [1]

**(b)** Small pies cost \$0.75 each to make.
Medium pies cost \$1.50 each to make.
Large pies cost \$2.25 each to make.
Represent these amounts in a $3 \times 1$ column matrix **N**. [1]

**(c)** Evaluate the matrix $\mathbf{T} = \mathbf{PN}$. [2]

**(d)** State what each of the elements of **T** represents. [1]

**(e)** The bakery sells each pie for 40% more than it costs to make.
One week they sold $\frac{6}{7}$ of each size of the apple pies and $\frac{4}{5}$ of each size of the cherry pies made that week.
The unsold pies were given away.
Calculate the total amount of profit the bakery made that week. [4]

[N18/II/3]

3. Yvette is a French tutor.

She offers sessions for basic and advanced students on weekdays and at weekends.

Each student has a 12-week block of sessions with one session per week.

The matrix **L** shows the number of students she tutors each week in one 12-week block.

$$\begin{array}{rcl} & \begin{array}{cc}\text{Basic} & \text{Advanced}\end{array} & \\ \mathbf{L} = & \begin{pmatrix} 8 & 5 \\ 2 & 4 \end{pmatrix} & \begin{array}{l}\text{Weekday}\\ \text{Weekend}\end{array} \end{array}$$

**(a)** Evaluate the matrix $\mathbf{M} = 12\mathbf{L}$. [1]

**(b)** Yvette charges \$40 for each basic session and \$65 for each advanced session.

Represent the session charges in a $2 \times 1$ column matrix **N**. [1]

**(c)** Evaluate the matrix $\mathbf{P} = \mathbf{MN}$. [1]

**(d)** State what the elements of **P** represent. [1]

**(e)** Yvette wants to attract more students, so in the next 12-week block she reduces her prices by 5%.

For this block of sessions, on weekdays she has 12 basic students and 6 advanced students.

On weekends she has 7 basic students and 3 advanced students.

Calculate the total amount of money she earns for this 12-week block of sessions. [3]

[N16/II/2]